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Studierichting: Informatica
Oefeningenreeks 5F nr. 1.6

$r = -4 \sin 3\theta$

$r' = -12 \cos 3\theta$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
r	0	$-$	0	$+$	0	$-$	0
r'	$-$	0	$+$	0	$-$	0	$+$
r	$\searrow$	$m(-4)$	$\nearrow$	$M(4)$	$\searrow$	$m(-4)$	$\nearrow$

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$$S_1 = \frac{1}{2}\int_0^{\frac{\pi}{3}} (-4\sin 3\theta)^2 d\theta$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{3}} 16\sin^2 3\theta d\theta$$

$$= 8\int_0^{\frac{\pi}{3}} \sin^2 3\theta d\theta$$

$$= 8\int_0^{\frac{\pi}{3}} \frac{1-\cos 6\theta}{2} d\theta$$

$$= \left[4\theta - \frac{2}{3}\sin 6\theta\right]_0^{\frac{\pi}{3}}$$

$$= \frac{4\pi}{3}$$

$$S = S_1 \cdot 3$$

$$= 4\pi$$

References